WEEK 2 – PRACTICAL

PROGRAM 1:

1. Try uname, lscpu, lsblk, ps, top commands after this program

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
        char cmd[100];

        printf("Enter Linux command: ");
        scanf("%s",cmd);
        pid_t pid = fork();
        if(pid == 0)
        {
                printf("Child PID : %d\n", getpid());
                execlp(cmd, cmd, NULL);
                perror("Execution Failed");
        }
        else if(pid > 0)
        {
                printf("Parent PID : %d\n",getpid());
                wait(NULL);
                perror("Execution Failed");
        }
        else
        {
                printf("Fork Failed\n");
        }
        return 0;
}
```

Execution:

```
ketana_sindhu@KetaZone:~$ ./file.c
Enter Linux command: ls
Parent PID : 1247
Child PID : 1248
DisplayProcessID.c  FirsFile     Fork.c   PID.c             S2Commands  WAIT       example.txt  ketana         pid       wait
File                FirstFile    HELLO.c  ParentAndChild.c  Sindhu      Wait.c     file         ketana_sindhu  shortcut  wait.c
File.c              FirstFile.c  KL.c     ReadUser.c        Sindhu.c    WeekTwo.c  file.c       name           sindhu
Execution: Success
ketana_sindhu@KetaZone:~$ |
```

Using uname : System info

```
ketana_sindhu@KetaZone:~$ uname -a
Linux KetaZone 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
ketana_sindhu@KetaZone:~$ |
```

Using lscpu : Computer architecture

```
ketana_sindhu@KetaZone:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          46 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   18
  On-line CPU(s) list:    0-17
Vendor ID:                GenuineIntel
  Model name:             Intel(R) Core(TM) Ultra 5 125H
    CPU family:           6
    Model:                170
    Thread(s) per core:   2
    Core(s) per socket:   9
    Socket(s):            1
    Stepping:             4
    BogoMIPS:             5990.39
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdts
                          cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
                          sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
                          bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
                          opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
                          clear serialize ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    432 KiB (9 instances)
  L1i:                    576 KiB (9 instances)
  L2:                     18 MiB (9 instances)
  L3:                     18 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-17
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Not affected
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
ketana_sindhu@KetaZone:~$ |
```

Using lsblk : Block storage devices

```
ketana_sindhu@KetaZone:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                                /
ketana_sindhu@KetaZone:~$ |
```

Using ps : Running process

```
ketana_sindhu@KetaZone:~$ ps
    PID TTY          TIME CMD
    638 pts/0    00:00:00 bash
    925 pts/0    00:00:00 ps
ketana_sindhu@KetaZone:~$ 
```

PROGRAM 2:

copying the contents from one file to another file

```
  GNU nano 8.7.1
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
        int src, dest;
        char buffer[1024];
        int bytes;

        src = open("input.txt", O_RDONLY);

        if(src < 0)
        {
                printf("Cannot open source file\n");
                return 1;
        }

        dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
        while((bytes = read(src,buffer, sizeof(buffer))) > 0)
        {
                write(dest,buffer,bytes);
        }
        close(src);
        close(dest);

        printf("File copied successfully.\n");

        return 0;
}
```

```
ketana_sindhu@KetaZone:~$ nano WeekTwoONE.c
ketana_sindhu@KetaZone:~$ gcc WeekTwoONE.c -o FILE.C
ketana_sindhu@KetaZone:~$ ./FILE.C
Cannot open source file
ketana_sindhu@KetaZone:~$ nano WeekTwoONE.c
ketana_sindhu@KetaZone:~$ 
```